Project Report

Project Title: Delivery Delay Analysis Using Power BI

[Problem Statement]

Delivery times for a food delivery service are being impacted by multiple factors such as peak-hour demand and adverse weather conditions. This impacts customer satisfaction and operational efficiency. The objective is to understand how delivery time varies with factors such as city, traffic, weather, and delivery practices, and use this analysis to improve SLA compliance and optimize operations.

[Result]

The analysis identified key delay drivers using Power BI dashboards. Metrics such as average delay, peak delay time slots, and zone-based performance were computed. Based on findings, actionable strategies were recommended to reduce late deliveries.

[Business Impact]

1. Improved understanding of delay patterns enabling better resource allocation.
2. Operational improvements leading to reduced delays and higher customer satisfaction.
3. Better preparedness for traffic/weather conditions resulting in fewer SLA breaches.

[Dataset & Assumptions]

Original dataset: 45,584 rows × 20 columns. After cleaning: 34,487 rows.
Key fields include city (zone), time_orderd, time_taken_min, weather_conditions, traffic_density, multiple_deliveries, and festival.
SLA (by city): Metropolitan = 27 min, Urban = 22 min, Semi-Urban = 49 min.
Delay calculation: Delay_Minutes = time_taken_min – SLA.

[Key Analysis & Insights]

- Overall delayed orders: 45.7%. Avg delivery time: 26.4 min.
- Highest delay during dinner slot (19:00–21:00): Avg +5 min, ~65% late.
- Urban zone is slightly worse than Metropolitan; Semi-Urban too small for conclusion.
- Weather: Cloudy & Fog show ~+3 min delay; Sunny is favorable (-3.7 min).
- Traffic: Jam + Fog/Cloudy = critical risk (>+10 min delay, >85% late).
- Multiple deliveries (>1) cause severe overruns (up to +18 min).

[Recommendations]

- Restrict batching during dinner peak; deploy extra riders between 19:00–21:00.
- Enable weather-aware ETA adjustments during Fog/Cloudy conditions.
- Optimize routing during high traffic and micro-zone assignments to reduce travel time.
- Introduce dynamic SLAs by zone × time slot for realistic planning.